Question	Answer	Marks	AOs		Guidance
14	$\frac{1}{2} - \sin 2x \cos x = \sin x \cos 2x$	**M1**	**3.1a**		
	$\sin 3x = \frac{1}{2}$ **oe**	**M1**	**2.1**	from compound angle formula allow sign errors only	*or* $4\sin^3 x - 3\sin x + ½ = 0$ **oe**
	$x = \frac{\pi}{18}$ and $x = \frac{5\pi}{18}$	**A1** **A1**	**3.2a** **1.1**	**A1** for each	0.17453… **A1** 0.87266… **A1** to 2 or more sf
	$\pm \int \left(\sin x \cos 2x - \left(\frac{1}{2} - \sin 2x \cos x \right) \right) \mathrm{d}x$ **oe**	**M1**	**1.1**	ignore limits	
	$\mathrm{F}[x] = -\frac{x}{2} - \frac{\cos 3x}{3}$	**A1**	**1.1**	allow the positive of this	$\pm\left(-\frac{4}{3}\cos^3 x + \cos x - \frac{x}{2}\right)$ **oe** or $\pm(-⅓\cos 2x \cos x + ⅓\sin x \sin 2x - ½x)$ **oe** for **A1**
	$\mathrm{F}[\frac{5\pi}{18}] - \mathrm{F}[\frac{\pi}{18}]$	**M1**	**1.1**	F[x] must be one of the correct forms	F[0.87266] – F[0.17453] for **M1**
	$\frac{\sqrt{3}}{3} - \frac{\pi}{9}$ or $\frac{3\sqrt{3}-\pi}{9}$ **cao**	**A1** **[8]**	**3.2a**		